Quick Exam Refresher – SAA-C03 Cheat Sheet

> **🎯 Complete last-minute study guide for the AWS Certified Solutions Architect – Associate (SAA-C03) exam. This page covers every major topic from all 4 domains in a quick, scannable format. Read top-to-bottom for a full course refresher.**

Note: Some content here may be outdated as AWS updates/removes services and features. So if you notice any issues, please contact @Rojan Sedhai.

🏛️ Domain 1: Design Secure Architectures (30%)

IAM – Identity & Access Management

- **Users** = individual people. **Groups** = collections of users. Cannot nest groups inside groups.
- **Roles** = the most secure way to grant permissions. Assumed by users, apps, or AWS services. Use roles for all EC2-to-service access and cross-account access — never hardcode keys.
- **Policies** = JSON documents (Allow/Deny). Attached to users, groups, or roles.
- **Principle of Least Privilege** = grant only the minimum permissions needed.
- **Root account**: set up MFA and billing alerts only. Never use for daily tasks.
- **IAM Identity Center (SSO)** = single sign-on for multiple AWS accounts.
- **Permission Boundaries** = max permissions an IAM entity can have, even if their policy allows more, used by IAM users or roles (not Groups).

- **Service Control Policies (SCPs)** = applied at the AWS Organization level to restrict what accounts can do. They do NOT grant permissions — they only restrict.

S3 Security

- **Bucket Policies** = resource-based JSON policies. Use for cross-account access or making a bucket public.
- **ACLs** = legacy, object-level. Prefer bucket policies for new designs.
- **Block Public Access** = account-level or bucket-level kill switch. Always enable unless intentionally public.
- **Pre-Signed URLs** = temporary access to private objects. Exam scenario: "User paid for premium video — how to let them watch without making bucket public?" → **Pre-Signed URL**.
- **S3 Access Points** = distinct entry points with their own policy (e.g., Finance Access Point → `/finance/` prefix only). Great for shared data lakes.
- **Encryption at rest:**

Type	Key Owner	Exam Keyword
SSE-S3	AWS	"AES-256", "Default", "No overhead"
SSE-KMS	AWS (KMS)	"Audit via CloudTrail", "Key rotation control"
SSE-C	Customer	"You hold keys", "HTTPS required"
Client-Side	Client	"Encrypt before upload"

- **Force HTTPS**: Bucket policy with `"aws:SecureTransport": "false"` → Deny.
- **MFA Delete** = requires MFA + Root user to permanently delete versions.
- **Object Lock (WORM)**:
  - **Governance Mode**: Admin can override with special permissions.
  - **Compliance Mode**: No one (not even root) can delete during retention period.

VPC Essentials

Concept	Key Detail
Public Subnet	Has a route to an Internet Gateway (IGW)
Private Subnet	No IGW route; uses NAT Gateway for outbound
NAT Gateway	AWS-managed, HA within one AZ. Charged per GB. Not free.
Security Groups	Stateful. Instance-level. Only inbound rules needed — return traffic auto-allowed.
NACLs	Stateless. Subnet-level. Must define BOTH inbound AND outbound rules.
VPC Peering	Connects two VPCs. NOT transitive (A↔B and B↔C ≠ A↔C).
Transit Gateway	Hub-and-spoke. Connects many VPCs and on-premises. Transitive routing supported.
PrivateLink	Private connectivity to AWS services without going over the internet. No VPC peering needed.
VPC Endpoints	**Gateway** = for S3 and DynamoDB (free). **Interface** = for most other services (charged).
Site-to-Site VPN	Encrypted tunnel over the internet. Fast to set up. Variable performance.
Direct Connect	Dedicated private physical connection. Consistent performance. Takes weeks to set up.

- **Bastion Host** = EC2 in public subnet used to SSH into private EC2 instances.
- **VPC Flow Logs** = capture info about IP traffic. Sent to S3 or CloudWatch Logs.

Key Security Services

Service	What It Does
AWS KMS	Manage encryption keys. Used by S3, EBS, RDS, etc.
AWS Secrets Manager	Store, rotate, and retrieve credentials/API keys automatically
AWS CloudTrail	Logs ALL API calls across your AWS account (who did what, when, from where)
AWS Config	Tracks resource configuration changes over time. Enables compliance checks.

Service	What It Does
Amazon GuardDuty	Continuously monitors AWS environments for threats. Uses ML, anomaly detection, and threat intelligence to analyze data sources like CloudTrail, VPC Flow Logs, and DNS logs, helping detect issues such as compromised credentials, unauthorized access, and malware.
Amazon Inspector	Automated vulnerability scanning for EC2 and container images in ECR and Lambda.
AWS Shield Standard	Always-on, free DDoS protection for all AWS customers
AWS Shield Advanced	Paid, 24/7 DDoS support, cost protection, near-real-time notifications
AWS WAF	Block web exploits (SQLi, XSS) at ALB, CloudFront, API Gateway
Amazon Macie	Discovers and protects sensitive data (PII, credentials) in S3 using ML
ACM	Free SSL/TLS certificates for use with ALB, CloudFront, API Gateway
AWS Organizations	Multi-account management. Consolidated billing. Apply SCPs.
AWS Control Tower	Sets up a secure, multi-account environment using best practices (Landing Zone)
AWS RAM	Share resources (subnets, Transit Gateways) across accounts
AWS Security Hub	Central dashboard for all security findings (GuardDuty, Inspector, Macie)

🌊 Domain 2: Design Resilient Architectures (26%)

High Availability vs Fault Tolerance vs DR

- **High Availability (HA)** = minimize downtime. May have brief interruption. Example: Multi-AZ RDS with auto-failover.
- **Fault Tolerance** = zero downtime, zero data loss. More complex and expensive.
- **RPO** = Recovery Point Objective — how much data loss you can tolerate.

- **RTO** = Recovery Time Objective — how long you can tolerate downtime.
- **DR Strategies (fastest/most expensive → slowest/cheapest):**
  1. **Multi-Site Active/Active** — zero RTO/RPO, full dual deployment
  2. **Warm Standby** — scaled-down live environment in secondary region
  3. **Pilot Light** — only core services running (e.g., DB replication active)
  4. **Backup & Restore** — cheapest, highest RTO/RPO

EC2 Essentials

Concept	Detail
AMI	Blueprint for EC2: OS, software, config. Can be region-specific or shared cross-account.
Instance Store	Physically attached storage. Fastest possible I/O. **Ephemeral** — data lost on stop/terminate.
EBS	Persistent block storage. Survives instance stop/restart. Attached to one AZ.
Placement Groups	Cluster = lowest latency, same rack. Spread = max isolation, different hardware. Partition = groups on separate partitions (for Kafka, HDFS).
EC2 Hibernate	Saves RAM to EBS. Instance resumes exactly where it left off.
User Data	Bootstrap scripts run at first launch. Used for software installation.
IMDSv2	Secure way for EC2 to query its own metadata. Requires session-based tokens.

Auto Scaling

- **Launch Template** = defines AMI, instance type, SG, key pair. Use instead of Launch Configuration (legacy).
- **Auto Scaling Group (ASG)** = automatically adds/removes EC2 instances.
- **Scaling Policies:**
  - **Target Tracking** = keep a metric at a target (e.g., CPU at 50%). Simplest.
  - **Step Scaling** = add/remove instances in steps based on alarm breach size.

    - **Scheduled Scaling** = scale based on known time patterns (e.g., every day at 8 AM).
    - **Predictive Scaling** = ML-based, forecasts future demand and pre-scales.
- **Cooldown Period** = time ASG waits after a scaling action before scaling again.

EC2 Pricing

Model	Savings	Use Case
On-Demand	Baseline	Unpredictable, short-term
Reserved (1yr)	~40%	Predictable, steady workloads
Reserved (3yr)	~60%+	Long-term commitment
Savings Plans	~66%	Flexible, applies to Lambda & Fargate too
Spot	~90%	Fault-tolerant, interruptible batch jobs
Dedicated Host	—	BYOL licensing, compliance needs

Elastic Load Balancing (ELB)

Type	Protocol	Use Case
ALB (Application)	HTTP/HTTPS/WebSocket	Path-based & host-based routing, microservices, Lambda targets
NLB (Network)	TCP/UDP/TLS	Ultra-low latency, static IP, high throughput
GLB (Gateway)	IP	Route traffic through 3rd-party security/inspection appliances

- **Stickiness (Session Affinity)** = ALB sends the same user to the same backend instance using a cookie.
- **Cross-Zone Load Balancing** = distributes traffic evenly across instances in ALL enabled AZs. Enabled by default for ALB, optional for NLB.
- **Connection Draining (Deregistration Delay)** = time ELB waits for in-flight requests to complete before removing an instance (default 300s).

RDS & Databases

Feature	Multi-AZ	Read Replica
Goal	High Availability	Read Scalability
Replication	Synchronous	Asynchronous
Readable?	❌ No	✅ Yes
Failover	Automatic (DNS flip)	Manual (must promote)

- **RDS Multi-AZ**: Standby in different AZ. Same endpoint — app needs no code changes on failover.
- **Read Replicas**: Up to 5 per instance (15 for Aurora). Can be cross-region (paid data transfer).
- **Aurora**: AWS-native. 6 copies across 3 AZs. Storage auto-scales to 128 TB. Up to 15 Read Replicas. 5x faster than MySQL.
- **Aurora Serverless**: Scales to zero. Great for intermittent/unpredictable workloads.
- **Aurora Global Database**: < 1 second cross-region replication. Promote secondary region in < 1 minute.
- **Aurora Backtrack (MySQL only)**: Rewind DB to any point in the past without restoring a snapshot. "CTRL+Z for your database."
- **RDS Proxy**: Connection pooler between Lambda/app and DB. Prevents connection overload. Reduces Aurora failover time by 66%.
- **RDS Custom**: For Oracle and SQL Server only. Gives OS/SSH access while still managed.
- **Encrypting an unencrypted RDS DB**: Snapshot → copy with encryption enabled → restore from encrypted snapshot.

Amazon ElastiCache

	Redis	Memcached
Persistence	✅ Yes	❌ No
Multi-AZ / HA	✅ Yes	❌ No

	Redis	Memcached
Data Types	Rich (Lists, Sets, Sorted Sets, Hashes)	Strings only
Best For	Leaderboards, pub/sub, sessions, durability	Simple caching, multi-threaded scale-out

- **Caching Patterns:**
  - **Lazy Loading** = cache on read miss. First request is slow; data can go stale.
  - **Write-Through** = update cache on every write. No stale data; wasted memory if data not read.
  - **TTL** = set expiry on keys to prevent staleness.

Amazon DynamoDB

- **Key-value + document** NoSQL. Single-digit ms at any scale. Fully managed (serverless).
- **Primary Key options:**
  - **Partition Key only** = must be unique per item.
  - **Partition Key + Sort Key (composite)** = PK groups items; SK sorts within the group.
- **Capacity modes:**
  - **Provisioned** = specify RCU/WCU. Use Auto-Scaling. Cheaper for predictable workloads.
  - **On-Demand** = auto-scales instantly. More expensive per request but no capacity planning.
- **Secondary Indexes:**
  - **LSI (Local)** = same partition key, different sort key. Created at table creation only. Supports strong consistency. 10 GB limit per partition.
  - **GSI (Global)** = any partition/sort key. Can be added anytime. Eventual consistency only. Has its own capacity.

- **DAX (DynamoDB Accelerator)** = in-memory cache for DynamoDB. Microsecond reads. Write-through.
- **DynamoDB Streams** = 24-hour ordered change log. Triggers Lambda on INSERT/UPDATE/DELETE.
- **Global Tables** = multi-region, multi-master (active-active). Requires Streams to be enabled.
- **TTL** = auto-delete items after a timestamp. Free — doesn't use WCU.
- **PITR** = Point-in-Time Recovery. Restore to any second in last 35 days.

Decoupling & Messaging

- **SQS** = message queue. Producers send, consumers pull. Messages persist until consumed.
  - **Standard Queue** = at-least-once delivery, best-effort ordering.
  - **FIFO Queue** = exactly-once, strict ordering. Up to 3,000 msg/sec with batching.
  - **Visibility Timeout** = time a message is "hidden" while being processed. If not deleted, it reappears.
  - **Dead Letter Queue (DLQ)** = messages that fail processing multiple times go here for inspection.
  - **Long Polling** = wait up to 20s for messages. Reduces empty responses and API costs.
- **SNS** = pub/sub. Push messages to many subscribers (SQS, Lambda, email, HTTP/S, SMS).
  - **Fan-out pattern**: SNS → multiple SQS queues. Decouple processing pipelines.
  - **SNS FIFO** = strict ordering + deduplication. Can only deliver to SQS FIFO.
- **EventBridge** = serverless event bus. Routes events from AWS services, SaaS, or custom sources to targets (Lambda, SQS, Step Functions). Use for event-driven architectures.

- **Step Functions** = visual workflow orchestrator for Lambda and other services. Supports retries, parallel execution, conditions.
- **Amazon MQ** = managed ActiveMQ/RabbitMQ. Use when migrating an existing app that uses standard AMQP/MQTT protocols and can't be refactored to use SQS/SNS.

Serverless & Containers

- **Lambda**: Serverless, event-driven compute. Max 15 min timeout. Auto-scales. Pay per invocation.
  - **Triggers**: S3, DynamoDB Streams, SQS, API Gateway, EventBridge, SNS, Kinesis, Cognito.
  - **Lambda@Edge** = run Lambda code at CloudFront Edge locations. Modify requests/responses.
- **ECS (Elastic Container Service)**: Managed container orchestration.
  - **EC2 Launch Type** = you manage the EC2 instances.
  - **Fargate Launch Type** = serverless. AWS manages the infrastructure.
- **EKS (Elastic Kubernetes Service)**: Managed Kubernetes. Use if you need Kubernetes API compatibility.
- **API Gateway**: Fully managed REST/HTTP/WebSocket API. Integrates with Lambda, HTTP backends, AWS services. Handles throttling, auth, caching.
- **Elastic Beanstalk**: PaaS. Deploy and manage web apps without managing EC2. Good for lift-and-shift web apps.

S3 Resilience & Storage

- S3 stores data across ≥3 AZs by default. 11 nines (99.999999999%) durability.
- **Versioning** = keep multiple object versions. Protects against overwrites and deletes.
- **Cross-Region Replication (CRR)** = DR, compliance, lower latency for global users.
- **Same-Region Replication (SRR)** = log aggregation, test environments.

    - Versioning must be enabled on both source AND destination.
- **Lifecycle Rules** = transition objects to cheaper tiers after N days, or delete after N days.
    - Transition waterfall: Standard → Standard-IA → Glacier Instant → Glacier Flexible → Deep Archive.
    - Cannot go backwards via lifecycle — must "Restore" from Glacier first.

S3 Storage Classes

Class	Availability	Retrieval	Use Case
Standard	99.99%	ms	Frequent access
Intelligent-Tiering	99.9%	ms	Unknown/unpredictable patterns
Standard-IA	99.9%	ms	Infrequent, instant access
One Zone-IA	99.5%	ms	Non-critical, easily reproducible. **Data lost if AZ fails.**
Glacier Instant	99%	ms	Archive needing quick retrieval
Glacier Flexible	99.9%	1 min – 12 hrs	Archive. Expedited/Standard/Bulk tiers.
Glacier Deep Archive	99.9%	12 – 48 hrs	Long-term compliance (7-10 years). Cheapest.

Route 53 & DNS

- **Routing Policies:**
    - **Simple** = single resource, no health checks.
    - **Weighted** = split traffic by percentage. A/B testing.
    - **Latency** = route to lowest-latency region.
    - **Geolocation** = route based on user's country/continent.
    - **Geoproximity** = route based on geographic distance (with bias). Requires Traffic Flow.

    - **Failover** = primary → secondary (health-check based active-passive).
    - **Multivalue** = returns multiple healthy IPs. Basic health checking. NOT a replacement for load balancers.
- **Health Checks** = monitor endpoints. Used by Failover, Weighted, Latency policies.
- **Alias Records** = free DNS record pointing to AWS resources (ELB, CloudFront, S3 website, etc.). Unlike CNAME, can point to a zone apex (e.g., `example.com`).

⚡ Domain 3: Design High-Performing Architectures (24%)

CloudFront (CDN)

- Caches static and dynamic content at **Edge Locations** worldwide to reduce latency.
- **Origins:** S3 bucket, ALB, EC2, S3 Static Website (treated as Custom Origin), on-prem server.
- **Origin Access Control (OAC)** = modern way to restrict access so only CloudFront can read from a private S3 bucket. Replaces legacy OAI.
- **Geo Restriction** = allow/block by country. For finer control (IPs, paths), use **AWS WAF**.
- **Cache Invalidation** = force CloudFront to remove cached content. First 1,000 paths/month free. Better architectural fix: **file versioning** (rename `image_v2.jpg`).
- **Price Classes** = reduce cost by limiting which Edge Locations serve your content (All → 200 → 100).
- **Exam trap**: CloudFront S3 Website endpoint = Custom Origin (not S3 Origin). OAC won't work; bucket must be public.

Global Accelerator vs CloudFront

	CloudFront	Global Accelerator
Purpose	Cache & serve content (CDN)	Improve global routing of non-cacheable traffic
Protocol	HTTP/HTTPS	TCP, UDP, HTTP (any)
Caching	✅ Yes	❌ No
IP	Varies (CDN IPs)	2 static Anycast IPs
Best For	Static/dynamic web, images, video	APIs, gaming, IoT, stateful apps

Storage Performance

Service	Best For
EBS gp3	General SSD. 3,000 IOPS baseline. IOPS & throughput configurable independently of size.
EBS io2 Block Express	Mission-critical databases. Up to 64,000 IOPS. Sub-millisecond latency.
EBS st1	Big data, logs, throughput-optimized HDD. NOT bootable.
EBS sc1	Coldest HDD. Cheapest block storage. Infrequent access.
Instance Store	Fastest possible I/O. Ephemeral — lost on stop/terminate.
EFS	Shared POSIX file system across multiple Linux EC2s. Auto-scales. Multi-AZ.
FSx for Windows	Shared Windows file system. SMB/NTFS. Active Directory integration.
FSx for Lustre	HPC, ML, video rendering. Integrates directly with S3 for data hydration.
FSx for NetApp ONTAP	Multi-protocol (NFS+SMB+iSCSI). Dedup, compression. Migrate NetApp on-prem.
FSx for OpenZFS	Migrate ZFS workloads. Microsecond latency. Snapshots and cloning.

- **EFS vs FSx for Windows**: Linux workloads → EFS. Windows/AD workloads → FSx for Windows.

- **EFS vs FSx for Lustre**: General home dirs → EFS. ML training on S3 data → FSx for Lustre.

Database Performance Quick Reference

DB	Type	Exam Keyword
RDS	Relational (SQL)	Managed SQL, compliance, complex queries
Aurora	Relational (SQL)	High performance, 6 copies, 15 replicas
DynamoDB	NoSQL (Key-Value)	Single-digit ms, serverless, billions of records
ElastiCache Redis	In-memory cache	Leaderboards, sessions, sub-ms reads
Redshift	Data Warehouse (OLAP)	Columnar, BI, analytics, petabyte scale
Athena	Serverless SQL on S3	Ad-hoc queries, pay per TB scanned
OpenSearch	Search + Log Analytics	Full-text search, Kibana dashboards
EMR	Big Data Clusters	Hadoop, Spark, Hive, MapReduce
Neptune	Graph DB	Social graphs, fraud detection, relationships
DocumentDB	Document DB	MongoDB-compatible
Keyspaces	Wide-column	Cassandra-compatible (CQL)
Timestream	Time-series	IoT sensor data, trillions of events/day
QLDB	Ledger DB	Immutable, cryptographically verifiable, centralized

Amazon Redshift

- OLAP (analytical), not OLTP (transactional). Columnar storage + MPP (Massively Parallel Processing).
- **Leader Node** = query planning. **Compute Nodes** = execution. **Slices** = parallel sub-partitions.
- **COPY command** = best way to load data. Parallelizes using MPP. Never use INSERT for bulk loads.

- **Redshift Spectrum** = query S3 data directly from Redshift without loading it. Joins S3 cold data with hot Redshift data.
- **Athena vs Redshift Spectrum**: Athena is ad-hoc with no cluster. Spectrum is for joining S3 data WITH existing Redshift data.
- **DR**: Automated snapshots every 8 hours. Manual snapshots retained until deleted. Copy snapshots cross-region.

Amazon Athena

- Serverless SQL on S3. Pay per TB scanned. Built on Presto.
- **Cost optimization (the Big Three):**
  1. **Partitioning** = sub-folders by date (e.g., `year=2023/month=01/`). Scans only relevant folder.
  2. **Columnar formats** = Parquet or ORC. Scans only requested columns. Convert with **AWS Glue**.
  3. **Compression** = Snappy, GZIP. Smaller files = less data scanned.
- **Federated Query** = query data in DynamoDB, RDS, on-prem via Lambda Data Source Connectors.

Real-Time Data

	Kinesis Data Streams	Kinesis Data Firehose
Latency	Real-time (~200ms)	Near real-time (~60s minimum)
Management	Manual shards (or On-Demand)	Fully serverless
Storage	24 hrs – 365 days retained	No storage — just moves data
Consumer	You write code (Lambda, KCL)	Managed delivery to S3/Redshift/OpenSearch
Exam Trigger	"Custom processing", "replay data", "real-time"	"Load to S3/Redshift", "convert to Parquet", "zero admin"

- **KDS Shards** = 1 shard = 1 MB/s write, 2 MB/s read.

- **Enhanced Fan-Out** = each consumer gets its own 2 MB/s. Prevents throttling across multiple consumers.
- **Firehose Destinations**: S3, Redshift (via S3), OpenSearch, Splunk, custom HTTP endpoint.
- **Amazon Managed Apache Flink** (fmr Kinesis Data Analytics) = SQL/Java on streaming data. Windowing functions (tumbling, sliding). Real-time aggregations.
- **Amazon MSK** = managed Kafka. Use only if app must use Kafka API. Otherwise prefer Kinesis.

AWS Glue

- Serverless ETL (Extract, Transform, Load). No infrastructure to manage.
- **Data Catalog** = centralized metadata store. Used by Athena and Redshift Spectrum.
- **Crawlers** = scan S3/RDS/DynamoDB, infer schema, populate the Data Catalog.
- **ETL Jobs** = Python or Scala (on Spark). Processes and transforms data.
- **Job Bookmarks** = tracks already-processed data so only new files are processed on next run. Keyword: "prevent reprocessing."
- **Glue DataBrew** = visual data cleaning for analysts (no code).

Amazon EMR

- Managed cluster for Hadoop, Spark, Hive, Presto on EC2.
- **Node types:**
  - **Master Node** = manages cluster. Must be On-Demand or Reserved.
  - **Core Node** = runs tasks AND stores HDFS data. Must be On-Demand or Reserved (data risk).
  - **Task Node** = runs tasks, no HDFS storage. Use **Spot Instances** here for cost savings.

- **EMRFS** = run EMR jobs reading/writing directly from S3 (persistent, not ephemeral like HDFS).

Networking Performance

- **Direct Connect** = dedicated physical private connection to AWS. Consistent latency. Takes weeks to provision. **Not encrypted by default** — add IPsec VPN on top for encryption.
- **Direct Connect Gateway** = use ONE DX connection to reach VPCs in multiple regions.
- **DX + VPN** = encrypted DX. Best of both: private path + encryption.
- **DX Resiliency**: Production → 2 connections at 2 different locations. Maximum → 4 total connections.
- **Site-to-Site VPN** = encrypted over internet. Quick to set up. Use as backup for DX.
- **AWS DataSync** = automated data transfer from on-prem to S3/EFS/FSx. Up to 10x faster than rsync. Preserves metadata. Use an Agent VM installed on-prem.
- **Snow Family:**

Device	Capacity	Use Case
Snowcone	8–14 TB	IoT, edge, backpack-portable
Snowball Edge	80–210 TB	DC migration, edge compute
Snowmobile	100 PB	Exabyte-scale (send a truck)

- **Rule of thumb**: If network transfer takes > 1 week, use Snowball instead.

AI/ML Services (Know What, Not How)

Service	Input → Output	Exam Keyword
SageMaker	Custom ML models	"Build/train/deploy ML", "Jupyter notebooks"
Rekognition	Image/Video → Analysis	"Facial recognition", "detect unsafe

Service	Input → Output	Exam Keyword
		content", "count people"
Transcribe	Audio → Text	"Speech-to-text", "subtitles", "meeting minutes"
Polly	Text → Audio	"Text-to-speech", "lifelike voices"
Translate	Text → Translated text	"Language translation", "localization"
Comprehend	Text → Insights	"Sentiment analysis", "NLP", "detect PII"
Textract	Scanned doc → Structured data	"OCR", "extract tables/forms from invoices"
Lex	Voice/Text → Chatbot	"Chatbot", "Alexa-powered conversational interface"
Kendra	Enterprise docs → Smart search	"Internal enterprise search", "search PDFs/SharePoint"
Forecast	Historical data → Future predictions	"Time-series forecasting", "predict sales/inventory"

- **Rekognition vs Textract**: Street sign photo → Rekognition. Scanned invoice → Textract.
- **Transcribe vs Polly**: Audio → Text (Transcribe). Text → Audio (Polly).
- **Kendra vs OpenSearch**: Internal document search (Kendra). App/log search engine (OpenSearch).

💸 Domain 4: Design Cost-Optimized Architectures (20%)

S3 Cost Optimization

- **S3 Intelligent-Tiering** = automatically moves objects between access tiers. No retrieval fees. Ideal when access patterns are unknown.
- **S3 Lifecycle Policies** = auto-transition to cheaper storage classes after N days.

- **S3 Select / Glacier Select** = filter data server-side before retrieving. Reduces bandwidth and cost.
- **Requester Pays** = the downloader (not the bucket owner) pays for data transfer. Use for public data sharing.
- **S3 Storage Lens** = organization-wide storage activity dashboard.

Compute Cost Optimization

- **Reserved Instances (RI)**: Standard RI = cheapest, locked to instance family/region. Convertible RI = can change instance type, cheaper savings.
- **Savings Plans**: Compute Savings Plans (flexible, covers EC2 + Lambda + Fargate). EC2 Savings Plans (locked to instance family).
- **Spot Instances**: 90% cheaper. Can be interrupted with 2-minute warning. Best for: batch jobs, CI/CD, stateless web servers, ML training.
- **Spot Fleets** = mix of Spot + On-Demand. Maintains target capacity even when Spot is interrupted.
- **AWS Compute Optimizer** = ML-based right-sizing recommendations for EC2, EBS, Lambda, ECS.
- **EC2 Auto Scaling** = scale IN during low demand to remove unnecessary instances.
- **Lambda** = pay only per invocation and duration. Zero cost when not running.
- **Fargate** = serverless containers. No idle EC2 costs.

Cost Management Tools

Tool	Purpose
AWS Cost Explorer	Visualize spending, forecast future costs, identify trends
AWS Budgets	Set cost/usage thresholds and get alerts via SNS or email
Cost & Usage Report (CUR)	Most detailed billing data. Delivered to S3. Query with Athena.
AWS Trusted Advisor	Checks for cost savings, security gaps, performance, fault tolerance

Tool	Purpose
AWS Compute Optimizer	Right-sizing recommendations using ML
AWS License Manager	Track and manage BYOL software licenses (SQL Server, Oracle)

Cost Optimization Patterns

- Use **CloudFront** to reduce data transfer costs — data served from edge avoids per-GB charges from EC2/S3.
- Use **Direct Connect** for high-volume data transfer — cheaper than internet egress per GB.
- Use **VPC Endpoints** for S3/DynamoDB — avoid NAT Gateway charges for internal traffic.
- Use **S3 lifecycle policies** to automatically archive old objects.
- Use **Aurora Serverless** or **DynamoDB On-Demand** for variable workloads to avoid paying for idle capacity.
- Use **Spot Instances** for Task Nodes in EMR (no HDFS data at risk).
- **RDS Read Replicas cross-region = paid** data transfer. Same-region = free.

🧠 Must-Know Concepts at a Glance

AWS Global Infrastructure

- **Regions** = isolated geographic areas. At least 3 AZs per Region.
- **Availability Zones (AZs)** = physically separate data centers. Low-latency connection between them. Deploy across multiple AZs for HA.
- **Edge Locations** = for CloudFront, Route 53, Lambda@Edge. More numerous than Regions.
- **Local Zones** = extend a Region to a metro area for lower latency.
- **Outposts** = AWS rack deployed in your data center. Extension of a public Region.

- **Wavelength** = AWS compute embedded in 5G networks for ultra-low-latency mobile apps.
- **Regional vs Global Services**: IAM, Route 53, CloudFront, WAF = Global. EC2, RDS, S3, VPC = Regional.

Shared Responsibility Model

- **AWS is responsible for**: Security **OF** the cloud — physical hardware, data centers, hypervisor, managed service infrastructure.
- **You are responsible for**: Security **IN** the cloud — OS patching, application security, IAM config, data encryption, network config.
- **Shared**: Patch management (you do OS, AWS does managed services), configuration management, training.

Well-Architected Framework – 6 Pillars

1. **Operational Excellence** – Automate operations, monitor, learn from failure.
2. **Security** – Least privilege, encryption at rest/transit, monitoring.
3. **Reliability** – Multi-AZ, backups, automatic recovery, horizontal scaling.
4. **Performance Efficiency** – Right instance types, serverless, caching, CDN.
5. **Cost Optimization** – Reserved/Spot, lifecycle policies, right-sizing, remove idle resources.
6. **Sustainability** – Minimize environmental impact, maximize utilization.

Migration Strategies (The "6 R's")

Strategy	Meaning
Retire	Decommission — no longer needed
Retain	Leave on-prem for now (compliance, dependency)
Rehost	Lift-and-shift to EC2. No changes.
Replatform	Lift, tinker, shift. Minor optimizations (e.g., move to RDS).
Repurchase	Move to SaaS (e.g., Salesforce, ServiceNow)

Strategy	Meaning
Refactor/Re-architect	Rebuild using cloud-native services (most expensive but most benefit)

Key Service Comparisons at a Glance

Scenario	Answer
Lambda overloading RDS with connections	**RDS Proxy**
Share one DX connection across multiple regions	**Direct Connect Gateway**
Real-time streaming with replay capability	**Kinesis Data Streams**
Load streaming data into S3 with no code	**Kinesis Data Firehose**
Query S3 data with SQL, no infrastructure	**Amazon Athena**
Analytics on joined S3 + Redshift data	**Redshift Spectrum**
Private S3 access through CloudFront	**OAC (Origin Access Control)**
Shared file system for Windows EC2	**FSx for Windows File Server**
Shared file system for Linux EC2	**Amazon EFS**
HPC / ML training on S3 data	**FSx for Lustre**
NoSQL for gaming leaderboards	**DynamoDB + DAX**
Encrypt Direct Connect traffic	**Site-to-Site VPN over DX**
Move 500 TB with slow internet	**AWS Snowball Edge**
Temporary access to private S3 object	**Pre-Signed URL**
Multi-account governance & billing	**AWS Organizations + SCPs**
Prevent accidental S3 deletion at org level	**SCP + MFA Delete + Versioning**
App needs to queue messages, strict order	**SQS FIFO**
Fan-out: one event to many consumers	**SNS → multiple SQS queues**
Migrate ActiveMQ broker to AWS	**Amazon MQ**
Detect threats in VPC Flow Logs	**Amazon GuardDuty**
Central view of all security findings	**AWS Security Hub**

⚠️ Common Exam Traps

- **Multi-AZ RDS ≠ read scaling** → use Read Replicas for reads. Multi-AZ is for HA only.
- **NAT Gateway is NOT free** → charges per GB processed.
- **S3 is object storage, not a file system** → use EFS for shared Linux file mounts.
- **CloudFront ≠ Global Accelerator** → CloudFront caches HTTP content; GA is for non-HTTP/stateful apps.
- **SQS Standard ≠ ordered** → use FIFO for strict ordering.
- **Security Groups are stateful; NACLs are stateless** → NACLs need explicit inbound AND outbound rules.
- **IAM Roles > IAM Users for service access** → never hardcode access keys in EC2 or Lambda.
- **Spot Instances can be interrupted** → don't use for critical, non-interruptible work.
- **Direct Connect is NOT encrypted by default** → add VPN over DX for encryption.
- **Aurora Backtrack = MySQL only** → doesn't work for PostgreSQL-compatible Aurora.
- **LSI must be created at table creation** → cannot add to existing table. GSI can be added anytime.
- **S3 Static Website endpoint = Custom Origin for CloudFront** → OAC won't work; bucket must be public.
- **SCPs don't grant permissions** → they only restrict the maximum permissions accounts can have.
- **Glacier Lifecycle: can't go backwards** → must Restore first; then copy to Standard if needed.
- **Reserved Instances in the wrong family** → Convertible RI lets you exchange; Standard RI locks you in.
- **Same-region RDS Read Replica data transfer = FREE; Cross-region = PAID**.

> 💡 **Exam Day Tips:**

> - Read every scenario question fully — AWS tests your ability to pick the MOST appropriate AND cost-effective solution.

> - When the question mentions "migrate existing X protocol" (AMQP, MQTT, Kafka), think managed compatibility service (MQ, MSK).

> - "Serverless + unpredictable traffic" → Lambda, Fargate, Aurora Serverless, DynamoDB On-Demand.

> - "Lowest latency globally" → CloudFront (HTTP) or Global Accelerator (TCP/UDP).

> - "Compliance + encryption + audit" → CloudTrail + KMS + Config.

> - When stuck: eliminate wrong answers first. AWS almost never recommends manually managing things that have a managed alternative.

📋 SAA-C03 Exam Guide – In-Scope Services Quick Reference

> One or two line descriptions for every AWS service listed in the official SAA-C03 Exam Guide appendix, organized by category.

Analytics

Service	Description
Amazon Athena	Serverless SQL query engine that runs queries directly on S3 data. Pay per TB scanned; no cluster to manage.
AWS Data Exchange	Marketplace to subscribe to and exchange third-party data sets in the cloud. Delivers data directly to S3.
AWS Data Pipeline	Managed ETL service to move and transform data between AWS compute and storage services on a schedule. Legacy service — prefer AWS Glue for new designs.

Service	Description
Amazon EMR	Managed big data cluster platform running Hadoop, Spark, Hive, and Presto on EC2. Use Spot Instances on Task Nodes for cost savings.
AWS Glue	Serverless ETL service with a Data Catalog (metadata store) used by Athena and Redshift Spectrum. Crawlers auto-discover schema.
Amazon Kinesis	Streaming data platform. **Data Streams** = real-time, retain up to 365 days, you write consumers. **Firehose** = fully managed delivery to S3/Redshift/OpenSearch.
AWS Lake Formation	Service to build, secure, and manage data lakes on S3. Simplifies ingestion, cataloging, and fine-grained access control.
Amazon MSK	Managed Apache Kafka service. Use only when your application requires the native Kafka API; otherwise prefer Kinesis.
Amazon OpenSearch Service	Managed Elasticsearch/OpenSearch cluster for full-text search, log analytics, and Kibana dashboards.
Amazon QuickSight	Serverless business intelligence (BI) tool for creating interactive dashboards and visualizations from AWS data sources.
Amazon Redshift	Columnar data warehouse (OLAP) for petabyte-scale analytics. Uses MPP for fast parallel query execution.

Application Integration

Service	Description
Amazon AppFlow	No-code integration service that transfers data between SaaS apps (Salesforce, Slack) and AWS services like S3 and Redshift.
AWS AppSync	Managed GraphQL API service that connects apps to data sources (DynamoDB, Lambda) with real-time and offline sync capabilities.
Amazon EventBridge	Serverless event bus that routes events from AWS services, SaaS apps, or custom sources to targets like Lambda or SQS. Core service for event-driven architectures.
Amazon MQ	Managed ActiveMQ and RabbitMQ broker. Use when migrating apps that use standard protocols (AMQP, MQTT) and can't be refactored to SQS/SNS.
Amazon SNS	Pub/sub messaging service. Pushes messages to multiple subscribers (SQS, Lambda, email, SMS). Use SNS → SQS for fan-out patterns.

Service	Description
Amazon SQS	Managed message queue. Producers send, consumers pull. Standard = at-least-once; FIFO = exactly-once with strict ordering.
AWS Step Functions	Visual serverless workflow orchestrator. Chains Lambda and other services with retries, branching, and parallel execution.

AWS Cost Management

Service	Description
AWS Budgets	Set custom cost or usage thresholds and receive alerts via SNS or email before overspending occurs.
AWS Cost and Usage Report (CUR)	Most granular billing data available; delivered to S3 and queryable with Athena for deep cost analysis.
AWS Cost Explorer	Visual tool to analyze spending trends, forecast future costs, and identify savings opportunities.
Savings Plans	Flexible commitment-based discount (up to 66%) in exchange for a consistent usage amount. Covers EC2, Lambda, and Fargate.

Compute

Service	Description
AWS Batch	Fully managed batch computing service. Automatically provisions EC2 or Spot Instances to run containerized batch jobs at any scale.
Amazon EC2	Virtual servers in the cloud. Choose instance type (compute/memory/storage optimized), pricing model (On-Demand, Reserved, Spot), and placement.
Amazon EC2 Auto Scaling	Automatically adds or removes EC2 instances based on demand using target tracking, step, scheduled, or predictive scaling policies.
AWS Elastic Beanstalk	PaaS that deploys and manages web apps (Java, Node.js, Python, etc.) without requiring EC2 management. Good for lift-and-shift web apps.
AWS Outposts	AWS-managed rack of hardware installed in your own data center. Extends AWS services (EC2, RDS, ECS) to on-premises environments.

Service	Description
AWS Serverless Application Repository	Catalog of pre-built serverless applications and components you can deploy directly into your AWS account.
VMware Cloud on AWS	Run VMware vSphere workloads natively on AWS bare-metal infrastructure. Used for migrating VMware environments without refactoring.
AWS Wavelength	Embeds AWS compute and storage within 5G carrier networks for ultra-low-latency mobile and edge applications.

Containers

Service	Description
Amazon ECR	Fully managed Docker container registry. Stores, versions, and secures container images for use with ECS and EKS.
Amazon ECS	AWS-native container orchestration service. Run containers on EC2 (you manage) or Fargate (serverless).
Amazon EKS	Managed Kubernetes service. Use when you need Kubernetes API compatibility or are migrating existing Kubernetes workloads.
Amazon ECS Anywhere / EKS Anywhere	Extensions of ECS and EKS that let you run containerized workloads on your own on-premises infrastructure using the same AWS control plane.
Amazon EKS Distro	The same Kubernetes distribution used by EKS, available for self-managed deployments on-premises or in other environments.

Database

Service	Description
Amazon Aurora	AWS-native relational DB. 6 copies across 3 AZs, up to 15 read replicas, storage auto-scales to 128 TB. 5× faster than MySQL.
Amazon Aurora Serverless	Aurora that automatically scales capacity to zero when idle. Ideal for intermittent or unpredictable workloads.

Service	Description
Amazon DocumentDB	MongoDB-compatible managed document database. Use when migrating MongoDB workloads without changing application code.
Amazon DynamoDB	Fully managed key-value and document NoSQL database. Single-digit millisecond performance at any scale; serverless.
Amazon ElastiCache	In-memory caching layer. Redis = HA, persistence, rich data types. Memcached = simpler, multi-threaded horizontal scaling.
Amazon Keyspaces	Serverless, managed Apache Cassandra-compatible wide-column database. Use CQL queries; scales automatically.
Amazon Neptune	Managed graph database. Supports Property Graph (Gremlin) and RDF (SPARQL). Best for social networks, fraud detection, and knowledge graphs.
Amazon QLDB	Immutable, cryptographically verifiable ledger database. Every change is recorded and tamper-proof. Centralized (not decentralized like blockchain).
Amazon RDS	Managed relational database service supporting MySQL, PostgreSQL, SQL Server, Oracle, and MariaDB. Handles backups, patching, and Multi-AZ.
Amazon Redshift	*(See Analytics above)*

Developer Tools

Service	Description
AWS X-Ray	Distributed tracing service that maps requests as they flow through microservices. Helps identify bottlenecks and errors in complex architectures.

Front-End Web and Mobile

Service	Description
AWS Amplify	Full-stack toolset for building and deploying web and mobile apps with built-in authentication, APIs, and hosting powered by AWS.
Amazon API Gateway	Fully managed service to create, publish, and secure REST, HTTP, and WebSocket APIs. Integrates natively with Lambda and other AWS

Service	Description
	services.
AWS Device Farm	Cloud-based testing service that runs your mobile and web apps on real physical devices hosted by AWS.
Amazon Pinpoint	Multi-channel marketing and messaging service (email, SMS, push notifications). Segment audiences and track engagement.

Machine Learning

Service	Description
Amazon Comprehend	NLP service for text analysis — extracts entities, sentiment, key phrases, and detects PII.
Amazon Forecast	Time-series forecasting service using ML. Input historical data to predict future demand, inventory, or resource needs.
Amazon Fraud Detector	ML-based fraud detection service. Builds models from your historical transaction data to flag suspicious activity in real time.
Amazon Kendra	Intelligent enterprise search service. Searches internal documents (PDFs, SharePoint, S3) using natural language understanding.
Amazon Lex	Build conversational chatbots using voice and text, powered by the same technology as Alexa.
Amazon Polly	Text-to-speech service. Converts written text to lifelike audio. Used for accessibility features and voice-enabled apps.
Amazon Rekognition	Image and video analysis service. Detects objects, faces, text, and unsafe content. Supports real-time face comparison.
Amazon SageMaker	End-to-end ML platform to build, train, and deploy custom machine learning models. Provides Jupyter notebooks and managed training infrastructure.
Amazon Textract	OCR-plus service that extracts text, tables, and forms from scanned documents and images. Goes beyond simple OCR to understand document structure.
Amazon Transcribe	Automatic speech recognition (ASR) — converts audio to text. Used for meeting transcripts, subtitles, and call center analytics.
Amazon Translate	Neural machine translation service for real-time or batch translation across 75+ languages.

Management and Governance

Service	Description
AWS Auto Scaling	Umbrella service that centrally manages Auto Scaling for EC2, ECS, DynamoDB, Aurora, and more via a single interface.
AWS CloudFormation	Infrastructure as Code (IaC) service. Define AWS resources in JSON/YAML templates and provision them repeatably as stacks.
AWS CloudTrail	Logs every API call made in your AWS account (who, what, when, from where). Essential for auditing and compliance.
Amazon CloudWatch	Centralized monitoring service for metrics, logs, alarms, and dashboards across all AWS services. Triggers Auto Scaling and Lambda.
AWS CLI	Command-line tool to interact with all AWS services via scripts or terminal. Useful for automation and administration.
AWS Compute Optimizer	Uses ML to analyze EC2, EBS, Lambda, and ECS usage and recommend right-sized resources to reduce cost and improve performance.
AWS Config	Tracks configuration changes to AWS resources over time. Enables compliance rules and auditing of resource history.
AWS Control Tower	Sets up and governs a secure, multi-account AWS environment (Landing Zone) using best practices. Sits on top of AWS Organizations.
AWS Health Dashboard	Provides personalized alerts about AWS service events and planned changes that may affect your specific resources.
AWS License Manager	Tracks and enforces BYOL (Bring Your Own License) software licenses (e.g., SQL Server, Oracle) across EC2 and on-premises.
Amazon Managed Grafana	Managed Grafana service for visualizing operational metrics from CloudWatch, Prometheus, and other data sources.
Amazon Managed Service for Prometheus	Managed time-series monitoring compatible with the Prometheus query language. Integrates with EKS and Grafana.
AWS Organizations	Multi-account management service. Enables consolidated billing, account grouping into OUs, and applying Service Control Policies (SCPs).

Service	Description
AWS Proton	Managed delivery service for containerized and serverless applications. Platform teams define infrastructure templates; developers self-serve.
AWS Service Catalog	Allows organizations to create and manage approved IT service portfolios (CloudFormation templates) for self-service provisioning.
AWS Systems Manager	Operational management hub for EC2 and on-premises servers. Includes Run Command, Parameter Store, Patch Manager, and Session Manager.
AWS Trusted Advisor	Automated checks across cost, security, performance, fault tolerance, and service limits. Some checks require Business/Enterprise support.
AWS Well-Architected Tool	Guided review of your workloads against the 6 pillars of the Well-Architected Framework. Identifies risks and improvement opportunities.

Media Services

Service	Description
Amazon Elastic Transcoder	Legacy media transcoding service. Converts video files stored in S3 to formats for playback on phones, tablets, and browsers.
Amazon Kinesis Video Streams	Ingests, stores, and processes live video streams from millions of devices. Integrates with Rekognition for real-time video analysis.

Migration and Transfer

Service	Description
AWS Application Discovery Service	Collects on-premises server inventory and usage data to help plan cloud migrations. Works with AWS Migration Hub.
AWS Application Migration Service (MGN)	Lift-and-shift migration service (formerly CloudEndure). Replicates servers to AWS with minimal downtime.
AWS DMS	Database Migration Service. Migrates databases to AWS with minimal downtime. Supports homogeneous (MySQL→RDS MySQL)

Service	Description
	and heterogeneous (Oracle→Aurora) migrations.
AWS DataSync	Automated, accelerated data transfer from on-premises NFS/SMB/HDFS storage to S3, EFS, or FSx. Up to 10× faster than rsync. Requires an agent VM on-prem.
AWS Migration Hub	Central dashboard to track the progress of application migrations from multiple AWS migration tools.
AWS Snow Family	Physical edge devices for offline data transfer. **Snowcone** (8–14 TB), **Snowball Edge** (80–210 TB), **Snowmobile** (100 PB). Use when internet transfer > 1 week.
AWS Transfer Family	Managed file transfer service supporting SFTP, FTPS, and FTP directly into and out of S3 or EFS. No server management required.

Networking and Content Delivery

Service	Description
AWS Client VPN	Managed OpenVPN-based VPN that allows remote users to securely access AWS resources and on-premises networks from any device.
Amazon CloudFront	Global CDN that caches content at Edge Locations worldwide. Reduces latency for static and dynamic content delivery.
AWS Direct Connect	Dedicated private physical connection between your data center and AWS. Consistent bandwidth and latency; takes weeks to provision. Not encrypted by default.
Elastic Load Balancing (ELB)	Distributes incoming traffic across multiple targets. ALB = Layer 7 (HTTP/HTTPS). NLB = Layer 4 (TCP/UDP). GLB = routes to inspection appliances.
AWS Global Accelerator	Uses AWS's global network backbone and 2 static Anycast IPs to route users to the nearest healthy endpoint. Best for non-cacheable TCP/UDP traffic.
AWS PrivateLink	Provides private connectivity from your VPC to AWS services or partner services without traversing the public internet. No VPC peering required.
Amazon Route 53	Managed DNS service with health checks and routing policies (Simple, Weighted, Latency, Geolocation, Failover, Multivalue). Also a domain registrar.

Service	Description
AWS Site-to-Site VPN	Encrypted IPsec tunnel between your on-premises network and AWS VPC over the public internet. Fast to set up; variable latency.
AWS Transit Gateway	Hub-and-spoke network transit hub that connects multiple VPCs and on-premises networks. Supports transitive routing (unlike VPC Peering).
Amazon VPC	Logically isolated virtual network in AWS. You control IP ranges, subnets, route tables, internet gateways, and security settings.

Security, Identity, and Compliance

Service	Description
AWS Artifact	Self-service portal to access AWS compliance reports (SOC, PCI, ISO) and manage compliance agreements.
AWS Audit Manager	Automates evidence collection for audits against frameworks like PCI DSS, HIPAA, and SOC 2.
AWS Certificate Manager (ACM)	Provisions and renews free SSL/TLS certificates for use with ALB, CloudFront, and API Gateway.
AWS CloudHSM	Dedicated hardware security module (HSM) for managing cryptographic keys. You have exclusive control — AWS cannot access your keys. More control than KMS.
Amazon Cognito	Adds user sign-up, sign-in, and access control to web and mobile apps. Supports social identity providers (Google, Facebook) and SAML 2.0.
Amazon Detective	Investigates security findings from GuardDuty using graph analysis of CloudTrail, VPC Flow Logs, and other data. Helps determine root cause.
AWS Directory Service	Managed Microsoft Active Directory in the cloud. Integrates with on-premises AD or provides a standalone AWS Managed Microsoft AD.
AWS Firewall Manager	Centrally configures and manages AWS WAF, Shield Advanced, and Network Firewall rules across multiple accounts in AWS Organizations.
Amazon GuardDuty	Continuous threat detection using ML and threat intelligence on CloudTrail, VPC Flow Logs, and DNS logs. Detects compromised

Service	Description
	credentials, malware, and unauthorized access.
AWS IAM Identity Center (SSO)	Centralized single sign-on for multiple AWS accounts and business applications. Replaces manual cross-account role switching.
AWS IAM	Controls who can do what in AWS. Users, groups, roles, and JSON policies. Always use roles for service-to-service access; never hardcode keys.
Amazon Inspector	Automated vulnerability scanning for EC2 instances and container images in ECR. Continuously monitors for CVEs and network exposure.
AWS KMS	Managed service to create and control encryption keys used by S3, EBS, RDS, and many other services. Integrates with CloudTrail for key usage auditing.
Amazon Macie	Uses ML to automatically discover, classify, and protect sensitive data (PII, credentials) stored in S3.
AWS Network Firewall	Managed stateful network firewall and intrusion detection service for VPCs. Provides fine-grained traffic filtering at the subnet level.
AWS RAM	Resource Access Manager. Share AWS resources (subnets, Transit Gateways, Resolver rules) across accounts without needing to duplicate them.
AWS Secrets Manager	Stores, rotates, and retrieves secrets (DB credentials, API keys) automatically. Lambda integration for rotation; audit via CloudTrail.
AWS Security Hub	Central aggregation point for security findings from GuardDuty, Inspector, Macie, and partner tools. Provides a unified compliance score.
AWS Shield	**Standard** = always-on, free DDoS protection for all AWS customers. **Advanced** = paid, 24/7 DDoS response team, cost protection, near-real-time notifications.
AWS WAF	Web Application Firewall. Blocks common exploits (SQLi, XSS, bad bots) at ALB, CloudFront, and API Gateway using rules and rate limiting.

Serverless

Service	Description
AWS AppSync	*(See Application Integration above)*
AWS Fargate	Serverless compute engine for ECS and EKS. Removes the need to provision or manage EC2 instances for containers. Pay per vCPU and memory used.
AWS Lambda	Serverless, event-driven compute. Runs code in response to triggers (S3, DynamoDB, API Gateway, etc.). Max 15 min runtime. Auto-scales automatically.

Storage

Service	Description
AWS Backup	Centralized, policy-driven backup service across EC2, EBS, RDS, DynamoDB, EFS, FSx, and Storage Gateway. Simplifies compliance with backup retention policies.
Amazon EBS	Persistent block storage attached to EC2. Survives instance stop/restart. Exists in a single AZ. Types: gp3 (general), io2 (high IOPS), st1/sc1 (HDD).
Amazon EFS	Managed, elastic NFS file system shared across multiple Linux EC2 instances. Multi-AZ, auto-scales, pay for what you use.
Amazon FSx	Managed file systems for specialized workloads. **Lustre** = HPC/ML on S3. **Windows File Server** = SMB/AD integration. **NetApp ONTAP** = multi-protocol. **OpenZFS** = ZFS migration.
Amazon S3	Object storage with 11 nines durability, stored across ≥3 AZs. Supports versioning, lifecycle rules, replication, and multiple storage classes (Standard → Deep Archive).
Amazon S3 Glacier	Low-cost archival storage within S3. **Instant Retrieval** = ms. **Flexible Retrieval** = 1 min–12 hrs. **Deep Archive** = 12–48 hrs. Cheapest long-term storage option.
AWS Storage Gateway	Hybrid storage bridge between on-premises applications and AWS storage. Modes: **S3 File Gateway** (NFS/SMB to S3), **Volume Gateway** (iSCSI block storage), **Tape Gateway** (virtual tape library).